



Pre-Session Note: Introduction to Data Analytics with Power BI

Welcome to the Course!

Welcome to the **Data Analytics using Power BI** course at Aptech Kubwa! This pre-session note prepares you for the first session by introducing Power BI, explaining its importance, comparing it with tools like Excel and Tableau, and outlining the course's major scope. Please complete the installation steps and review this note before the first class to ensure a smooth start.

1. Installing Power BI Desktop

- **Definition:** Power BI Desktop is a free application from Microsoft for creating and analyzing data visualizations and reports.
- **Steps to Install:**
 - i. Visit [Microsoft Power BI Desktop Download](#).
 - ii. Click “Download” to get the installer for Windows (64-bit recommended).
 - iii. Run the installer and follow the prompts to install.
 - iv. Open Power BI Desktop and sign in with a Microsoft account (free or organizational).
 - v. Verify installation by checking the interface (Home, View, Modeling tabs).
- **System Requirements:**
 - Windows 10 or later.
 - At least 4 GB RAM (8 GB recommended).
 - 2 GB free disk space.
- **Example:** After installation, you’ll see a welcome screen with options like “Get Data” to import datasets.
- **Pre-Session Task:** Install Power BI Desktop and open it to confirm it’s working. Bring a laptop with the software installed to the first session.

2. What is Power BI?

- **Definition:** Power BI is a Microsoft business intelligence (BI) tool that enables users to connect to data sources, transform data, create interactive reports and dashboards, and share insights for decision-making.
- **Key Components:**
 - **Power BI Desktop:** For building reports and dashboards.
 - **Power BI Service:** Cloud-based platform for sharing and collaborating on reports.
 - **Power BI Mobile:** Apps for viewing reports on mobile devices.
- **Example:** A retail manager uses Power BI to visualize sales data from multiple stores, identifying top-selling products in a dashboard.
- **Scenario:** A school wants to analyze student performance across subjects. Power BI connects to their Excel grade sheets, transforming raw data into charts showing trends in student scores.

3. Why Use Power BI?

- **Purpose:** Power BI simplifies data analysis by offering intuitive tools for data integration, transformation, visualization, and sharing, enabling data-driven decisions.
- **Benefits:**
 - **Ease of Use:** Drag-and-drop interface for creating visuals without coding.
 - **Multiple Data Sources:** Connects to Excel, databases (e.g., SQL Server), and cloud services (e.g., Google Analytics).
 - **Interactive Dashboards:** Allows users to filter and drill down into data dynamically.
 - **Real-Time Insights:** Supports live data connections for up-to-date analysis.
 - **Cost-Effective:** Power BI Desktop is free, with affordable Pro licenses for advanced features.
- **Use Case:** A logistics company uses Power BI to track delivery times across regions, creating a dashboard that highlights delays and optimizes routes, improving efficiency.

4. Power BI vs. Excel vs. Tableau

- **Comparison Overview:**

Feature/Tool	Power BI	Excel	Tableau
Purpose	Business intelligence, interactive dashboards, and data sharing	Spreadsheet-based data analysis and basic visualization	Advanced data visualization and analytics
Data Sources	Hundreds (Excel, databases, cloud services)	Limited (Excel files, some databases)	Many (similar to Power BI)
Ease of Use	Beginner-friendly with drag-and-drop	Familiar to most but manual for complex tasks	Steeper learning curve
Visualization	Interactive dashboards with slicers	Basic charts (pivot charts, limited interactivity)	Highly customizable visuals
Data Volume	Handles large datasets efficiently	Limited by spreadsheet size	Handles large datasets well
Sharing	Power BI Service for cloud sharing	Manual (email, shared drives)	Tableau Server/Online (paid)
Cost	Free Desktop, Pro at ~\$10/user/month	Included in Microsoft 365	Higher cost (~\$70/user/month for Creator)

- **Example:**
 - **Excel:** A small business tracks monthly expenses in a pivot table but struggles with large datasets or sharing.
 - **Power BI:** The same business imports the Excel file into Power BI, creating an interactive dashboard shared via Power BI Service.
 - **Tableau:** A data analyst uses Tableau for complex visualizations but requires more training to build dashboards.
- **Scenario:** A marketing team needs to analyze campaign performance. Excel is used for

quick calculations, Power BI for interactive dashboards shared with stakeholders, and Tableau for advanced statistical visualizations by data experts.

5. Major Scope of the Course

Based on the Aptech curriculum (from the provided document), this course equips you with practical skills in Power BI for data analytics. By the end, you will be able to:

- **Import, Transform, and Clean Data:** Connect to various sources (e.g., Excel, databases) and use Power Query to ensure data accuracy.
 - *Example:* Cleaning a sales dataset by removing duplicates and standardizing formats.
- **Create Interactive Reports and Dashboards:** Build visuals (charts, tables) for business insights.
 - *Use Case:* A dashboard showing regional sales trends for a retail chain.
- **Apply Data Modeling:** Establish relationships between datasets for deeper analysis.
 - *Example:* Linking customer and sales tables to analyze purchasing patterns.
- **Use DAX (Data Analysis Expressions):** Perform calculations and aggregations.
 - *Example:* Calculating year-over-year sales growth.
- **Optimize Report Performance:** Set up data refresh schedules for real-time insights.
 - *Use Case:* Automating daily sales updates for a dashboard.
- **Share and Collaborate:** Use Power BI Service to share reports and embed them in platforms.
 - *Example:* Embedding a report in a company's intranet for team access.
- **Course Duration:** 40 online hours (likely 20 sessions of 2 hours each) + 12 self-study hours.
- **Resources:** Access eBooks, practice tests, assignments, and references via OnlineVarsity (as per the document).

Pre-Session Preparation

- **Task:** Install Power BI Desktop and test it by opening the application.
- **Self-Study:** Watch a 5-minute video on "What is Power BI?" (e.g., [Microsoft's Power BI Intro](#) – instructor to provide link).
- **Bring to Class:** A laptop with Power BI installed and a Microsoft account for signing in.
- **Reflection Question:** Think of a dataset you work with (e.g., sales, grades, expenses).

How could Power BI help analyze it?

Additional Resources

- Microsoft Power BI Documentation: <https://learn.microsoft.com/en-us/power-bi/>
- Comparison Guide: [Power BI vs. Tableau](#)
- Installation Guide: [Power BI Desktop Setup](#)